

Arabic-Centric Preference Optimization for Efficient Dialect-Sensitive Machine Translation: A Seq2Seq Approach

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Abstract

Arabic machine translation remains challenging because Modern Standard Arabic and dialectal varieties such as Egyptian Colloquial Arabic differ substantially in morphology, lexical choice, register, and everyday usage. We introduce a compact Arabic-centric encoder-decoder model pretrained from scratch with multilingual Seq2Seq training and contrastive cross-lingual alignment over Arabic, English, Egyptian Arabic, and 111 additional languages. We then apply Direct Preference Optimization (DPO) using preference pairs targeting Arabic-specific translation errors. On matched FLORES-200 language sets, our model outperforms the similarly sized NLLB-200-distilled-600M baseline by 3.82 average spBLEU for $xx \rightarrow \text{English}$, 5.76 for $xx \rightarrow \text{MSA}$, and 9.02 for $xx \rightarrow \text{ECA}$. DPO further improves $\text{MSA} \rightarrow \text{English}$ and $\text{ECA} \rightarrow \text{English}$ by approximately +2 spBLEU over the supervised fine-tuned checkpoint, while the final model remains competitive with substantially larger general-purpose LLMs.

1 Introduction

Arabic machine translation remains challenging because Modern Standard Arabic (MSA) and dialectal varieties such as Egyptian Colloquial Arabic (ECA) differ substantially in morphology, lexical choice, register, and everyday usage. MSA is used in formal writing and institutional communication, while ECA dominates everyday communication, media, and culturally specific expression. Recent Arabic-focused MT research increasingly points in the same direction: compact specialized models can outperform much larger general systems, multi-dialect training matters under constrained resources, and standard automatic metrics often fail to capture dialect fidelity, cultural appropriateness, and variety-specific lexical choice (Hennara et al., 2025; Alabdullah et al., 2025; Sibae et al., 2025; Nowshin et al., 2026; Magdy et al., 2026). In other

words, Arabic MT benefits less from generic scale alone than from architectures, supervision, data design, and evaluation that are explicitly designed for the language’s structure and sociolinguistic reality.

Machine translation in real-world deployment still depends primarily on encoder-decoder systems because they remain the most practical architecture for conditional generation at scale. In multilingual machine translation, encoder-decoder models continue to offer an effective balance of quality, efficiency, and controllability, especially for the long tail of language pairs where compute budgets and serving constraints matter (NLLB Team, 2022). Recent work reinforces this deployment logic, showing that translation systems which preserve an NMT-style decoder can retain strong quality while substantially reducing inference cost relative to decoder-heavy alternatives (Luo et al., 2025).

Yet the recent alignment wave in Natural Language Processing has only partially reached this architecture. While reinforcement learning methods such as Group Relative Policy Optimization (GRPO) have produced strong results for decoder-only LLMs in machine translation (Shao et al., 2024; Guo et al., 2025; He et al., 2025; Feng et al., 2025; Yang et al., 2025), evidence for encoder-decoder MT is still limited. The latest Seq2Seq study paints a mixed picture. Although reference-free GRPO improves many languages, Arabic remains a clear boundary case, with weak or negative gains attributed to reward-calibration limitations rather than optimization failure (Garcia-Estrada et al., 2026). This is particularly relevant for Arabic MT, where morphology and lexical ambiguity make translation quality difficult to compress into a single language-agnostic reward.

In this work, we introduce an Arabic-centric encoder-decoder machine translation model centered around Modern Standard Arabic and Egyptian Colloquial Arabic. Although the model re-

mains competitive beyond its focal directions, its design is optimized for the linguistic conditions in which generic multilingual systems are weakest: Arabic morphology, dialect-sensitive lexical choice, and everyday Egyptian usage. We apply Direct Preference Optimization (DPO) on curated preference pairs targeting linguistic phenomena that automated metrics often flatten, allowing the model to learn distinctions in adequacy and dialectal naturalness that are especially important in Arabic translation (Rafailov et al., 2023; Cui et al., 2025; Wang et al., 2025; Yang et al., 2026).

This design choice is deliberate. Rather than relying on reward-driven Reinforcement Learning (RL) with scalar quality signals that recent work has shown to be brittle for fine-grained MT ranking (Yang et al., 2026; Garcia-Estrada et al., 2026), we use preference optimization to encode the kinds of pairwise judgments that matter most for Arabic and Egyptian Arabic. The result is a deployment-oriented system that combines the serving advantages of encoder-decoder MT with Arabic-specific modeling decisions, and that, in our deployment tests, is lightweight enough to run on a single NVIDIA RTX 4000 Quadro GPU under Triton Inference Server while still delivering its strongest gains in Arabic-focused translation.

Contributions. Our contributions are as follows:

- We introduce a compact Arabic-centric encoder-decoder machine translation model focused on **MSA** and **ECA**, while maintaining competitive multilingual performance across non-Arabic language directions.
- We apply Direct Preference Optimization (DPO) using grounded preference pairs targeting Arabic-specific translation errors, including morphology, lexical ambiguity, register mismatch, and dialectal naturalness.
- We evaluate on FLORES-200, WMT24++, and NTREX (NLLB Team, 2022; Deutsch et al., 2025; Federmann et al., 2022). On matched FLORES-200 language sets, our model outperforms the similarly sized NLLB-200-distilled-600M baseline by 3.82, 5.76, and 9.02 average spBLEU for xx→English, xx→MSA, and xx→ECA, respectively.
- We demonstrate that our model is practical for real-world deployment, it can be served on a single NVIDIA RTX 4000 Quadro GPU using

Triton Inference Server, showing that high-quality Arabic MT can be achieved without relying on very large decoder-only LLMs.

2 Related Work

Seq2Seq multilingual machine translation. Despite the recent success of decoder-only LLMs, production machine translation has historically been dominated by encoder-decoder architectures because they directly condition generation on a source sentence and provide a strong efficiency-quality trade-off. Early multilingual pretraining work such as mBART showed that denoising pretraining of a complete sequence-to-sequence model can improve supervised, unsupervised, and document-level MT, especially in lower-resource settings (Liu et al., 2020). NLLB-200 further scaled this direction to more than 200 languages and remains one of the strongest open encoder-decoder baselines for multilingual MT (NLLB Team, 2022). Recent work has also argued that translation systems do not need to abandon encoder-decoder structure in order to benefit from LLM-era modeling improvements, since encoder-decoder or encoder-enhanced systems can retain strong translation quality while reducing inference cost relative to decoder-heavy alternatives (Luo et al., 2025). We follow this deployment-oriented line of work: rather than scaling a decoder-only LLM, we specialize a compact encoder-decoder model for Arabic and Egyptian Arabic while preserving competitive multilingual behavior outside the focal language directions.

Arabic and dialectal Arabic machine translation. Arabic MT remains difficult because Arabic is not a single homogeneous translation target. Modern Standard Arabic is used in formal writing and institutional communication, while dialectal varieties dominate everyday use, with Egyptian Colloquial Arabic being one of the most widely used and culturally influential varieties. Recent Arabic-focused systems have revisited the value of specialization. Mutarjim, for example, demonstrates that a small Arabic-English translation model can rival much larger LLMs through targeted data curation and training (Hennara et al., 2025). Work on dialectal-to-MSA translation shows that fine-tuned and quantized models can outperform zero-shot prompting for Egyptian, Levantine, and Gulf dialects, and that joint multi-dialect training can be more effective than separate dialect-specific models under constrained resources (Alabdullah et al.,

2025). In parallel, Egyptian-focused language modeling work such as Nile-Chat highlights the importance of modeling Egyptian Arabic in both Arabic and Latin scripts (Shang et al., 2025). LoRA-tuned LLM normalization has also been shown to improve the downstream translation of Arabic vernaculars (Alam and Anastasopoulos, 2025). Our model differs from these efforts by targeting deployable encoder-decoder MT for MSA and ECA, combining compact sequence-to-sequence modeling with Arabic-focused preference optimization rather than relying only on prompting, generic multilingual transfer, or model scale.

Arabic translation resources and evaluation.

Recent work has expanded Arabic MT resources beyond MSA and high-resource settings. ATHAR provides a 66K-sample Classical Arabic-English dataset (Mohammed and Khalil, 2025), while TEDxTN targets Tunisian Arabic-English speech translation with code-switching and regional variation (Bougares et al., 2025). Most relevant to dialectal MT, Alexandria introduces a large-scale human-translated benchmark covering multiple Arabic dialects, countries, and domains, making it an important resource for evaluating dialect-sensitive translation (El Mekki et al., 2026). More broadly, FLORES-200, NTREX, and WMT24++ provide multilingual and multi-domain MT evaluation coverage (NLLB Team, 2022; Federmann et al., 2022; Deutsch et al., 2025). However, dialect-sensitive Arabic MT remains difficult to evaluate with surface metrics alone, motivating our use of multiple benchmarks and preference data targeting Arabic-specific error categories (Magdy et al., 2026).

Reinforcement learning and preference optimization for MT. Sequence-level optimization for MT predates the current LLM alignment wave. Policy-gradient methods such as MIXER and minimum-risk training were introduced to reduce exposure bias and optimize sequence-level objectives more directly than token-level maximum likelihood (Ranzato et al., 2016; Shen et al., 2016). Modern RLHF later popularized Proximal Policy Optimization-based alignment with learned reward models (Schulman et al., 2017; Ouyang et al., 2022), while GRPO removed the value model and became a more memory-efficient post-training method (Shao et al., 2024; Guo et al., 2025). In MT, recent GRPO-style work has mainly targeted decoder-only LLMs: R1-T1 uses RL to elicit translation reasoning patterns (He et al., 2025), MT-R1-

Zero combines rule and metric rewards for LLM-based translation (Feng et al., 2025), and MERIT applies reward-guided optimization to Chinese-centric low-resource MT using language-specific token prefixing and semantic alignment rewards (Lu et al., 2026). In the encoder-decoder setting, recent work applies GRPO to NLLB-200 with reference-free LaBSE and COMET-Kiwi rewards and shows broad gains across many languages, but also identifies Arabic as a boundary case where GRPO provides weak or unstable improvements due to limited headroom, root-pattern morphology, and reward-model calibration gaps (Garcia-Estrada et al., 2026).

Direct Preference Optimization for Arabic MT.

Reference-free reward models such as LaBSE and COMET-Kiwi are attractive because they remove the need for target references during fine-tuning. However, scalar quality estimators can struggle to rank fine-grained translation differences, especially when adequacy, register, morphology, and dialectal naturalness interact. GRRM addresses this issue by replacing independent scalar scoring with group-relative reward modeling for MT, arguing that candidate groups should be compared jointly rather than scored in isolation (Yang et al., 2026). Preference optimization offers a complementary path. DPO removes the need for a separate reward model and RL sampling loop by directly optimizing pairwise preferences (Rafailov et al., 2023). MT-specific variants such as CRPO and M²PO further show that preference optimization can be adapted to translation by improving preference-pair selection, incorporating confidence or reward signals, and using richer multi-pair/multi-perspective supervision (Cui et al., 2025; Wang et al., 2025). For encoder-decoder MT, TWA uses span-level MQM error annotations and outperforms sequence-level DPO trained on the same data (Zhang et al., 2025). Because comparable span-level Arabic annotations are unavailable, we instead use sequence-level preference pairs. Our model adopts this preference-optimization direction because our target errors are not merely generic adequacy errors: they include Arabic morphology, lexical ambiguity, MSA-ECA register mismatch, dialectal naturalness, and culturally appropriate phrasing. These are precisely the types of distinctions that are easier to express as pairwise translation preferences than as a single language-agnostic scalar reward.

3 Data Collection

3.1 Pretraining and Fine-tuning Data

We trained our model using multilingual parallel corpus designed to preserve broad translation competence while increasing coverage of Arabic and Egyptian Colloquial Arabic. For the initial training stage, we follow the contrastive multilingual training direction introduced in mRASP2 (Pan et al., 2021). This is especially suitable for our setting because Egyptian Arabic is underrepresented in public parallel corpora, and many useful transfer paths may not appear as direct Egyptian Arabic parallel directions.

We construct the pretraining and supervised fine-tuning pool from several parallel resources, mainly through public multilingual parallel data from [liboaccn/nmt-parallel-corpus](#) as the main public source of multilingual parallel data. To strengthen the low-resource dialectal component, we supplement the public data with a private Egyptian Arabic parallel corpus. This corpus was professionally translated from English, French, and Chinese into Egyptian Arabic by human annotators. The private ECA data is important because publicly available dialectal Arabic parallel corpora remain limited, and generic multilingual corpora often underrepresent the informal phrasing, lexical choices, and register shifts required for natural Egyptian Arabic translation.

The resulting training mixture is therefore designed with two goals: maintaining general multilingual translation ability across many language directions, and increasing the density and quality of Arabic/ECA supervision. This allows our model to remain competitive outside Arabic while specializing in the directions where generic multilingual systems most often struggle.

3.2 Preference Data for Direct Preference Optimization

After supervised training, we observed that the translation model was already capable in terms of adequacy, but its outputs were often overly literal, stiff, or too close to formal written Arabic. This was especially problematic for Egyptian Arabic, where natural translation often requires dialectal phrasing, idiomatic wording, and register-sensitive lexical choices. We therefore construct a DPO dataset to target these specific weaknesses rather than relying only on additional parallel fine-tuning.

The DPO data is constructed from 20-language

multi-parallel records generated using Gemini 3 Flash. For each selected translation direction, the Gemini translation is used as the preferred candidate, while the corresponding output of our supervised fine-tuned (SFT) checkpoint is used as the rejected candidate; generation is conditioned on a requested domain and a fixed difficulty rubric, with Gemini providing a textual rationale for the linguistic difficulty of each sentence. The prompts are designed around failure modes observed in the SFT model’s outputs, including literal translation, incorrect register, unnatural dialectal phrasing, lexical ambiguity, and domain-specific terminology. We also include examples from specialized fields such as medicine and law, where incorrect or overly generic phrase choices can significantly harm translation quality.

Each generated instance is organized as a preference example, where the preferred translation better satisfies adequacy, fluency, register, and dialectal naturalness than the rejected alternative. To improve coverage and maintain general MT behavior, we also incorporate examples derived from the FLORES-200 development set (NLLB Team, 2022). These examples are used only for preference-data construction and are kept separate from the final evaluation splits used in our reported benchmark results. The final preference dataset contains approximately 80K examples, centered around Arabic and Egyptian Arabic directions while retaining multilingual diversity.

Because Egyptian Arabic is comparatively low-resource and harder to evaluate automatically, we manually reviewed 10% of the English–Egyptian Arabic parallel pairs to confirm semantic similarity between pairs and as a sanity check of the LLM-generated outputs. The complete generation prompt, output schema, language list, and preference-pair construction procedure are provided in Appendix E.

Annotation and filtering details. The DPO preference data is filtered to remove malformed pairs, duplicated examples, and cases where the preferred and rejected translations are semantically indistinguishable. For the Egyptian Arabic portion, annotators check adequacy, fluency, register, and dialectal naturalness. In addition to manually reviewing 10% of the English–Egyptian Arabic preference pairs, we recruited a qualified annotator with formal academic training in English language and pedagogy, holding a degree from the Department of English

at the Faculty of Education, to conduct an independent preference evaluation. The annotator was presented with 100 English–Egyptian Arabic samples, each containing the English source sentence together with the chosen and rejected translations, and was asked to identify which candidate provided the better Egyptian Arabic translation of the English source. The results of this evaluation are reported in Table 1.

Annotator preference	Percentage
Chosen translation preferred	76%
Rejected translation preferred	16%
Tie / No clear preference	8%

Table 1: Human preference evaluation on 100 randomly sampled English–Egyptian Arabic translation pairs.

4 Methodology

4.1 Contrastive Multilingual Pretraining

We pretrain our model with a multilingual sequence-to-sequence objective with the contrastive learning task introduced in mRASP2 (Pan et al., 2021). The goal of this stage is not only to learn conditional translation, but also to reduce the representation gap between semantically equivalent sentences across languages. This is especially important for our setting. Our aim was to retain broad multilingual competence while becoming substantially stronger on Arabic-centered directions.

Given a parallel pair (x^i, x^j) from languages L_i and L_j , the standard MT objective maximizes the likelihood of the target sentence conditioned on the source sentence:

$$\mathcal{L}_{\text{mt}} = - \sum_{(x^i, x^j) \in \mathcal{D}} \log p_{\theta}(x^j | x^i). \quad (1)$$

Following mRASP2, we compute a sentence-level representation $\mathcal{R}(x)$ by average-pooling the encoder hidden states of sentence x . For each true parallel pair (x^i, x^j) , the two sentences form a positive pair, while non-aligned sentences from the same mini-batch serve as negatives. We define:

$$s(a, b) = \frac{\text{sim}(\mathcal{R}(a), \mathcal{R}(b))}{\tau}, \quad (2)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is a temperature parameter. The contrastive objective is then:

$$\mathcal{L}_{\text{ctl}} = - \sum_{(x^i, x^j) \in \mathcal{D}} \log \frac{\exp(s(x^i, x^j))}{\sum_{y \in \mathcal{B}} \exp(s(x^i, y))}, \quad (3)$$

where $\mathcal{B}$ is the set of candidate sentences in the mini-batch.

The final pretraining objective combines token-level translation supervision with sentence-level cross-lingual alignment:

$$\mathcal{L}_{\text{pretrain}} = \mathcal{L}_{\text{mt}} + \lambda |s| \mathcal{L}_{\text{ctl}}, \quad (4)$$

where λ balances the two losses and $|s|$ scales the sentence-level contrastive term to the token-level MT loss, following mRASP2.

This stage gives our model a multilingual alignment backbone before Arabic-specific preference optimization.

4.2 Direct Preference Optimization for Arabic-Centered MT

After supervised training, we apply Direct Preference Optimization to improve the translation behavior of our SFT model (Rafailov et al., 2023). We choose DPO because the remaining errors are not primarily failures of basic adequacy. The model is already capable, but its outputs can be overly literal, overly formal, or insufficiently natural for Egyptian Arabic. These errors are difficult to capture with a single scalar reward, particularly for Arabic where morphology, register, dialectal phrasing, and lexical ambiguity interact. DPO instead allows us to optimize directly from pairwise translation preferences.

Each preference example is represented as (x, y^+, y^-) , where x is the source sentence with the target-language condition, y^+ is the preferred translation, and y^- is the rejected translation. For an encoder–decoder policy π_{θ} , the sequence log-probability is computed over the target tokens:

$$\log \pi_{\theta}(y | x) = \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t | y_{<t}, x). \quad (5)$$

To keep the DPO objective compact, we define the policy preference margin:

$$\Delta_{\theta} = \log \pi_{\theta}(y^+ | x) - \log \pi_{\theta}(y^- | x), \quad (6)$$

and the corresponding reference margin:

$$\Delta_{\text{ref}} = \log \pi_{\text{ref}}(y^+ | x) - \log \pi_{\text{ref}}(y^- | x). \quad (7)$$

DPO optimizes the model by increasing the preferred translation margin relative to the fixed reference policy:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{\mathcal{D}_{\text{pref}}} [\log \sigma(\beta(\Delta_{\theta} - \Delta_{\text{ref}}))], \quad (8)$$

where σ is the logistic sigmoid and β controls the strength of the preference update relative to the reference model. Intuitively, the model is rewarded when it assigns the preferred translation a larger likelihood margin than the rejected translation, without requiring a separately trained reward model or an online reinforcement learning loop.

A key implementation detail is that DPO is applied parameter-efficiently using LoRA (Hu et al., 2022) following (Garcia-Estrada et al., 2026) assumption. We update only low-rank adapter parameters while keeping the underlying encoder-decoder backbone fixed, which preserves the model’s multilingual translation competence and reduces training cost. We apply LoRA to the query and value projection matrices of the attention layers.

This optimization stage is designed to shift the SFT model from merely correct translation toward preferred translation. For Arabic and Egyptian Arabic, that distinction is crucial: a translation can be semantically adequate but still sound too literal, too formal, or unnatural for the intended dialect. By training on curated preference pairs, DPO directly targets these fine-grained distinctions while retaining the efficiency and controllability of the encoder-decoder architecture.

5 Experimental Setup

5.1 Model Architecture

Our model is a compact encoder-decoder machine translation model with approximately 560M parameters. It uses an 18 layers encoder-18 layers decoder architecture rather than a decoder-only architecture, allowing generation to be explicitly conditioned on the encoded source sentence.

5.2 Language Control Tokens

We used explicit language-control tokens to indicate the desired target language during training and inference. We add special tokens for the supported target languages, including an Egyptian Arabic token (`__arz__`) to distinguish Egyptian Arabic from Modern Standard Arabic. These tokens are used consistently during pretraining, supervised fine-tuning, DPO, and evaluation, allowing the same Seq2Seq model to translate into Arabic, Egyptian Arabic, English, and the remaining supported languages. The complete token list is provided in Appendix B, Table 16.

5.3 Pretraining Configuration

The pretraining corpus contains approximately 2B sentence pairs and back-translated examples. More than 500M examples involve English-Arabic directions, and approximately 100M examples involve Egyptian Arabic (`__arz__`) directions. Training is bidirectional, each parallel pair is used in both source-to-target and target-to-source directions, allowing the model to learn translation into and out of Arabic, English, and Egyptian Arabic.

We follow the optimization setup of mRASP2. The model is trained with Adam, using a learning rate of 3×10^{-4} , polynomial decay scheduling, and 10K warmup steps. We use $\epsilon = 10^{-6}$, $\beta_2 = 0.98$, gradient clipping with maximum norm 5.0, and contrastive loss weight $\lambda = 1.0$.

5.4 Preference Optimization

After supervised training, we further optimize SFT model using DPO. DPO training is performed on approximately 80K preference examples for 5 epochs using a single NVIDIA H100 GPU. The preference data is centered on Arabic and Egyptian Arabic translation behavior, with examples targeting literal translation, register mismatch, dialectal naturalness, lexical ambiguity, and domain-specific terminology.

We apply DPO using LoRA adapters on the attention projection layers. The LoRA rank is set to $r = 32$, with $\alpha = 64$ and dropout 0.05. The DPO temperature parameter is set to $\beta = 0.2$, and gradients are clipped with maximum norm 1.0.

5.5 Evaluation Benchmarks

We evaluate on three multilingual MT benchmarks: FLORES-200, WMT24++, and NTREX. FLORES-200 is used to measure broad multilingual generalization, WMT24++ provides multi-domain evaluation across diverse languages and dialects, and NTREX evaluates news-domain translation quality. We report results separately for Arabic-focused directions and general multilingual directions to distinguish specialization from overall multilingual competence.

5.6 Baselines

We evaluate **NLLB-200-distilled-600M** (NLLB Team, 2022), a multilingual encoder-decoder MT model of approximately 600M parameters, as a controlled same-scale baseline for our approximately 560M-parameter model. We use the public `facebook/nllb-200-distilled-600M`

checkpoint. For $xx \rightarrow \text{English}$, we also compare against **Qwen/Qwen3-Next-80B-A3B-Instruct** (Qwen Team, 2025), **openai/gpt-oss-120b** (OpenAI, 2025), and **Gemma4-E2B**, **Gemma4-E4B**, and **Gemma4-26B-A4B** (Google DeepMind, 2026); the same Gemma family is used for translation into MSA and ECA. Gemma was selected because it achieved the strongest spBLEU relative to model size among the general-purpose, non-MT-specialized LLMs we evaluated on FLORES-200, while its multiple scales support quality-size comparisons.

5.7 Evaluation Metrics

We use spBLEU as the automatic evaluation metric for all reported benchmark results. Scores are computed consistently across systems using the same references and preprocessing pipeline for each benchmark. We use spBLEU because it is the standard metric for FLORES-style multilingual MT evaluation and provides a consistent basis for comparing our model against larger multilingual systems. spBLEU combines BLEU (Papineni et al., 2002) with SentencePiece tokenization (Kudo and Richardson, 2018), following FLORES-style evaluation (Goyal et al., 2022) and consistent BLEU reporting practices (Post, 2018).

Because automatic metrics may not adequately capture dialectal naturalness and other fine-grained translation properties, we additionally conduct a human evaluation of English-to-Egyptian-Arabic conversations generated by the SFT and DPO models. For each English conversation, the annotator is shown the two corresponding translations under neutral labels and evaluates each translation independently according to three criteria:

- **Egyptian Naturalness** : how naturally the translation expresses the source content using Egyptian-Arabic vocabulary, phrasing, and register.
- **Semantic Similarity** : how accurately and completely the translation preserves the meaning of the original English conversation, without introducing, omitting, or altering information.
- **Grammatical Correctness** : how grammatically and structurally well formed the translation is.

Each criterion is scored independently using the six-point ordinal scale shown in Table 2.

Score	Rating	Description
5	Excellent	Fully correct and natural.
4	Good	Minor issues only.
3	Acceptable	Some noticeable issues.
2	Poor	Several important issues.
1	Very Poor	Major problems.
0	Incorrect	Incorrect or unusable.

Table 2: Human-evaluation scoring rubric used for Egyptian naturalness, semantic similarity, and grammatical correctness.

The same rubric is applied to both systems, enabling a direct comparison of the effects of DPO on dialectal naturalness, meaning preservation, and grammatical quality.

5.8 Decoding and Evaluation Protocol

All systems use the same source sentences, references, and spBLEU pipeline within each benchmark. For our model, we use deterministic 5-beam decoding, no sampling, a maximum generation length of 1022 new tokens, and a length penalty of -0.5 , with the target language enforced through the decoder beginning-of-sequence token. NLLB uses the official FLORES-200 language codes and corresponding forced beginning-of-sequence token with the same beam size, length penalty, and no sampling. For LLM baselines, we use a fixed translation prompt and parse only the generated translation.

6 Results

We provide representative qualitative examples of Arabic-specific improvements after DPO in Appendix C, focusing on register, dialectal naturalness, and literal-translation errors.

6.1 Headline Results and Post-training Gains

We first summarize the main Arabic-centered findings across FLORES-200, NTREX, and WMT24++. Table 3 reports DPO gains over the supervised fine-tuned (SFT) checkpoint, Table 4 summarizes the strongest Arabic-centered comparisons, and Table 5 reports benchmark-level averages. Detailed per-language results are provided in Appendix A. Scores are spBLEU, rounded to two decimal places.

Direction	SFT	DPO	Gain
MSA→English	41.94	43.91	+1.97
ECA→English	27.75	29.80	+2.05

Table 3: DPO results on FLORES-200 Arabic-centered xx→English directions.

Benchmark	Direction	Our Model	Best baseline	Diff.
FLORES-200	MSA→English	43.91	45.62	-1.71
FLORES-200	ECA→English	29.80	30.45	-0.65
NTREX	MSA→English	37.75	37.43	+0.32
WMT24++	ECA→English	24.91	24.49	+0.42
FLORES-200	English→MSA	41.87	43.66	-1.79
FLORES-200	English→ECA	30.02	29.80	+0.22
WMT24++	English→ECA	26.03	25.72	+0.31

Table 4: Arabic and Egyptian Arabic comparisons against the strongest reported baseline in each setting, including NLLB-200-distilled-600M and the evaluated LLMs. Positive differences indicate that our model is higher.

Benchmark	Direction	Our model	NLLB-600M	Gemma4-26B
FLORES-200	xx→en	42.05	38.23	45.04
FLORES-200	xx→MSA	29.98	24.22	32.52
FLORES-200	xx→ECA	25.28	16.26	23.72
NTREX	xx→en	39.39	—	40.90
WMT24++	xx→en	34.85	—	37.81
WMT24++	xx→ECA	23.74	—	24.34

Table 5: Average spBLEU scores. NLLB-200-distilled-600M was evaluated on FLORES-200 only; averages use the same 29 language directions.

Observations. On matched 29-language FLORES-200 sets, our model exceeds NLLB-200-distilled-600M by 3.82 average spBLEU for xx→English, 5.76 for xx→MSA, and 9.02 for xx→ECA. The gains are +7.39 for English→MSA and +5.17 for English→ECA, while NLLB is 0.65 higher on the individual ECA→English direction. DPO additionally improves our SFT model from 41.94 to 43.91 on MSA→English and from 27.75 to 29.80 on ECA→English. Against the larger LLM baselines, our model remains competitive and achieves the strongest English→ECA result.

6.2 Human Judgment Results

To complement the automatic evaluation, we recruited a professional annotator with a degree in English from the Faculty of Education to conduct an independent human evaluation of the SFT and DPO translations. Table 6 presents the mean human ratings assigned to the SFT and DPO translations across 100 English-to-Egyptian-Arabic conversations. Scores range from 0 to 5, with higher scores indicating better quality.

Criterion	SFT	DPO	Gain
Egyptian naturalness	4.12	4.37	+0.25
Semantic similarity	3.72	4.10	+0.38
Grammatical correctness	3.17	3.68	+0.51

Table 6: Mean human-evaluation scores for the SFT and DPO translations across 100 conversations. Scores range from 0 to 5. Gains are calculated as DPO minus SFT.

The DPO model received higher mean ratings than the SFT model across all three evaluation criteria. The largest improvement was observed in grammatical correctness (+0.51), followed by semantic similarity (+0.38) and Egyptian naturalness (+0.25). These results indicate that DPO improved the structural correctness and meaning preservation of the translations while also producing more natural Egyptian-Arabic output.

6.3 Translation into Arabic Varieties

We next evaluate translation into Modern Standard Arabic (xx→MSA) and Egyptian Arabic (xx→ECA). This setting directly tests the main specialization target of our model. The per-language tables for this setting are reported in Appendix A, Tables 12, 13, 14, and 15.

Observations. On FLORES-200, our model outperforms NLLB-200-distilled-600M on every source direction shown in Tables 12 and 13, with average margins of +5.76 spBLEU for xx→MSA and +9.02 for xx→ECA. It also remains competitive with the larger Gemma models and obtains the strongest English-to-ECA result on WMT24++.

6.4 Post-training Ablations

6.4.1 LORA settings

In addition to the final DPO configuration, we experimented with alternative LoRA settings. Table 7 summarizes the tested configurations. Increasing the adapter capacity to rank 64 with $\alpha = 128$ caused the model to hallucinate more frequently and deviate substantially from the reference model. In contrast, reducing the adapter capacity to rank 8 with $\alpha = 16$ produced outputs that did not differ significantly from the supervised model, either qualitatively or quantitatively. Through these experiments, we found that rank 32 with $\alpha = 64$ provided the best trade-off between adaptation strength and preservation of the reference model behavior.

Run	Rank	Alpha	Dropout	Beta
Low-rank DPO	8	16	0.05	0.2
Final DPO	32	64	0.05	0.2
High-rank DPO	64	128	0.05	0.2

Table 7: LoRA configuration ablations for DPO.

6.4.2 Group Relative Policy Optimization

We also attempted GRPO as an exploratory post-training method, but it led to negative gains in our setting. We used MetricX as the reward model because the source sentence, generated hypothesis, and reference translation were all available during training. For each source sentence, we sampled 16 rollouts and optimized the model with a learning rate of 2×10^{-7} . Despite this setup, GRPO did not provide a useful learning signal for our Arabic-centered translation objective. The MetricX reward did not separate the sampled outputs strongly enough, especially when the differences between candidates involved register, dialectal naturalness, or Arabic-specific lexical choices. As a result, the model did not receive sufficiently fine-grained feedback to improve Egyptian Arabic and Modern Standard Arabic translation quality. We also found alternative automatic rewards less suitable for our goal, since they did not adequately capture the Arabic-specific phenomena targeted in this work. We therefore report GRPO as a negative result and use DPO as the final post-training method.

6.4.3 DPO data size

We also study the effect of preference-data scale on Arabic-centered post-training. As shown in Table 8, using 20K preference examples already improves over the supervised model for both Modern Standard Arabic and Egyptian Arabic. Increasing the preference dataset to 80K examples leads to larger gains, improving MSA→English from 41.94 to 43.91 spBLEU and ECA→English from 27.75 to 29.80 spBLEU. These results suggest that the DPO stage benefits from additional Arabic-centered preference coverage, especially for Egyptian Arabic, where naturalness and register-sensitive lexical choices require more diverse supervision.

DPO data size	MSA→en	ECA→en
SFT only	41.94	27.75
20K	42.18	28.32
80K	43.91	29.80

Table 8: DPO data-size ablation.

7 Limitations

Our evaluation is based on automatic spBLEU, which is useful for consistent benchmark comparison but does not fully capture dialect fidelity, register, or cultural appropriateness. This is especially important for Egyptian Arabic, where a translation can be semantically adequate but still sound unnatural or overly formal. Future work should include human evaluation by native speakers of Egyptian Arabic and Modern Standard Arabic.

8 Conclusion

We present an Arabic-centric Seq2Seq machine translation model for MSA and ECA while preserving multilingual translation competence across 111 additional languages. Beyond supervised training, we apply DPO using curated preference pairs targeting Arabic-specific errors, including literal translation, register mismatch, dialectal naturalness, lexical ambiguity, and domain-specific terminology. On matched FLORES-200 sets, our model outperforms the similarly sized NLLB-200-distilled-600M baseline by 3.82, 5.76, and 9.02 average spBLEU for xx→English, xx→MSA, and xx→ECA, respectively, while DPO adds approximately two spBLEU on both MSA→English and ECA→English. These findings show that careful data design, language control, and preference optimization can produce an efficient Arabic-focused MT system without relying only on model scale.

References

- Abdullah Alabdullah, Lifeng Han, and Chenghua Lin. 2025. [Advancing dialectal arabic to modern standard arabic machine translation](#). *arXiv preprint arXiv:2507.20301*, arXiv:2507.20301.
- Md Mahfuz Ibn Alam and Antonios Anastasopoulos. 2025. [Large language models as a normalizer for transliteration and dialectal translation](#). In *Proceedings of the 12th Workshop on NLP for Similar Languages, Varieties and Dialects*, pages 39–67. Association for Computational Linguistics.
- Fethi Bougares, Salima Mdhaffar, Haroun Elleuch, and Yannick Est‘eve. 2025. [TEDxTN: A three-way](#)

- speech translation corpus for code-switched tunisian arabic–english. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 278–287, Suzhou, China. Association for Computational Linguistics.
- Guofeng Cui, Pichao Wang, Yang Liu, Zemian Ke, Zhu Liu, and Vimal Bhat. 2025. **Crpo: Confidence-reward driven preference optimization for machine translation**. *arXiv preprint arXiv:2501.13927*, arXiv:2501.13927.
- Daniel Deutsch, Eleftheria Briakou, Isaac Caswell, Mara Finkelstein, Rebecca Galor, Juraj Juraska, Geza Kovacs, Alison Lui, Ricardo Rei, Jason Riesa, Shruti Rijhwani, Parker Riley, Elizabeth Salesky, Firas Trabelsi, Stephanie Winkler, Biao Zhang, and Markus Freitag. 2025. **Wmt24++: Expanding the language coverage of wmt24 to 55 languages dialects**. *arXiv preprint arXiv:2502.12404*, arXiv:2502.12404.
- Abdellah El Mekki, Samar M. Magdy, Houdaifa Atou, Ruwa AbuHweidi, Baraah Qawasmeh, Omer Nacar, Thikra Al-hibiri, Razan Saadie, Hamzah Alsayadi, Nadia Ghezaiel Hammouda, Alshima Alkhazimi, Aya Hamod, Al-Yas Al-Ghafri, Wesam El-Sayed, Asila Al sharji, Mohamad Ballout, Anas Belfathi, Karim Ghaddar, Serry Sibae, and 28 others. 2026. **Alexandria: A multi-domain dialectal arabic machine translation dataset for culturally inclusive and linguistically diverse llms**. *arXiv preprint arXiv:2601.13099*, arXiv:2601.13099.
- Christian Federmann, Tom Kocmi, and Ying Xin. 2022. **NTREX-128 – news test references for MT evaluation of 128 languages**. In *Proceedings of the First Workshop on Scaling Up Multilingual Evaluation*, pages 21–24, Online. Association for Computational Linguistics.
- Zhaopeng Feng, Shaosheng Cao, Jiahua Ren, Jiayuan Su, Ruizhe Chen, Yan Zhang, Zhe Xu, Yao Hu, Jian Wu, and Zuozhu Liu. 2025. **Mt-r1-zero: Advancing llm-based machine translation via r1-zero-like reinforcement learning**. *arXiv preprint arXiv:2504.10160*, arXiv:2504.10160.
- Ernesto Garcia-Estrada, Carlos Escolano, and Jos'e A. R. Fonollosa. 2026. **Reference-free reinforcement learning fine-tuning for mt: A seq2seq perspective**. *arXiv preprint arXiv:2605.15976*, arXiv:2605.15976.
- Google DeepMind. 2026. **Gemma 4 model card**. Google AI for Developers.
- Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc'Aurelio Ranzato, Francisco Guzmán, and Angela Fan. 2022. **The FLORES-101 evaluation benchmark for low-resource and multilingual machine translation**. *Transactions of the Association for Computational Linguistics*, 10:522–538.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shitong Ma, Peiyi Wang, Xiao Bi, and 1 others. 2025. **Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning**. *arXiv preprint arXiv:2501.12948*, arXiv:2501.12948.
- Mingui He, Yilun Liu, Shimin Tao, Yuanchang Luo, Hongyong Zeng, Chang Su, Li Zhang, Hongxia Ma, Daimeng Wei, Weibin Meng, Hao Yang, Boxing Chen, and Osamu Yoshie. 2025. **R1-t1: Fully incentivizing translation capability in llms via reasoning learning**. *arXiv preprint arXiv:2502.19735*, arXiv:2502.19735.
- Khalil Hennara, Muhammad Hreden, Mohamed Motaism Hamed, Zeina Aldallal, Sara Chrouf, and Safwan AlModhayan. 2025. **Mutarjim: Advancing bidirectional arabic-english translation with a small language model**. *arXiv preprint arXiv:2505.17894*, arXiv:2505.17894.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. **LoRA: Low-rank adaptation of large language models**. In *Proceedings of the International Conference on Learning Representations*.
- Taku Kudo and John Richardson. 2018. **SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing**. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 66–71. Association for Computational Linguistics.
- Yinhan Liu, Jiatao Gu, Naman Goyal, Xian Li, Sergey Edunov, Marjan Ghazvininejad, Mike Lewis, and Luke Zettlemoyer. 2020. **Multilingual denoising pre-training for neural machine translation**. *arXiv preprint arXiv:2001.08210*, arXiv:2001.08210.
- Zhixiang Lu, Chong Zhang, Chenyu Xue, Angelos Stefanidis, Chong Li, Jionglong Su, and Zhengyong Jiang. 2026. **MERIT: Multilingual expert-reward informed tuning for chinese-centric low-resource machine translation**. *arXiv preprint arXiv:2604.04839*, arXiv:2604.04839.
- Yingfeng Luo, Tong Zheng, Yongyu Mu, Bei Li, Qinghong Zhang, Yongqi Gao, Ziqiang Xu, Peinan Feng, Xiaoqian Liu, Tong Xiao, and Jingbo Zhu. 2025. **Beyond decoder-only: Large language models can be good encoders for machine translation**. *arXiv preprint arXiv:2503.06594*, arXiv:2503.06594.
- Samar M. Magdy, Fakhreddin Alwajih, Abdellah El Mekki, Wesam El-Sayed, and Muhammad Abdul-Mageed. 2026. **Lqm: Linguistically motivated multi-dimensional quality metrics for machine translation**. *arXiv preprint arXiv:2604.18490*, arXiv:2604.18490.
- Mohammed Sabry Mohammed and Mohammed Khalil. 2025. **ATHAR: A high-quality and diverse dataset for classical arabic to english translation**. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 97–106, Suzhou, China. Association for Computational Linguistics.

- James Cross Onur Çelebi Maha Elbayad Kenneth Heafield Kevin Heffernan Elahe Kalbassi Janice Lam Daniel Licht Jean Maillard Anna Sun Skyler Wang Guillaume Wenzek Al Youngblood Bapi Akula Loic Barrault Gabriel Mejia Gonzalez Prangthip Hansanti John Hoffman Semarley Jarrett Kaushik Ram Sadagopan Dirk Rowe Shannon Spruit Chau Tran Pierre Andrews Necip Fazil Ayan Shruti Bhosale Sergey Edunov Angela Fan Cynthia Gao Vedanuj Goswami Francisco Guzmán Philipp Koehn Alexandre Mourachko Christophe Ropers Safiyyah Saleem Holger Schwenk Jeff Wang NLLB Team, Marta R. Costa-jussà. 2022. [No language left behind: Scaling human-centered machine translation](#).
- Afroza Nowshin, Prithweeraaj Acharjee Porag, Haziq Jeelani, and Fayeeg Jeelani Syed. 2026. [Context-aware dialectal arabic machine translation with interactive region and register selection](#). *arXiv preprint arXiv:2604.06456*, arXiv:2604.06456.
- OpenAI. 2025. [gpt-oss-120b & gpt-oss-20b model card](#). *Preprint*, arXiv:2508.10925.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. [Training language models to follow instructions with human feedback](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744.
- Xiao Pan, Mingxuan Wang, Liwei Wu, and Lei Li. 2021. [Contrastive learning for many-to-many multilingual neural machine translation](#). *arXiv preprint arXiv:2105.09501*, arXiv:2105.09501.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. [BLEU: a method for automatic evaluation of machine translation](#). In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA. Association for Computational Linguistics.
- Matt Post. 2018. [A call for clarity in reporting BLEU scores](#). In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191, Brussels, Belgium. Association for Computational Linguistics.
- Qwen Team. 2025. [Qwen3 technical report](#). *Preprint*, arXiv:2505.09388.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. [Direct preference optimization: Your language model is secretly a reward model](#). *arXiv preprint arXiv:2305.18290*, arXiv:2305.18290.
- Marc’Aurelio Ranzato, Sumit Chopra, Michael Auli, and Wojciech Zaremba. 2016. [Sequence level training with recurrent neural networks](#). In *Proceedings of the International Conference on Learning Representations*.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. [Proximal policy optimization algorithms](#). *arXiv preprint arXiv:1707.06347*, arXiv:1707.06347.
- Guokan Shang, Hadi Abdine, Ahmad Chamma, Amr Mohamed, Mohamed Anwar, Abdelaziz Bounhar, Omar El Herraoui, Preslav Nakov, Michalis Vazirgiannis, and Eric P. Xing. 2025. [Nile-chat: Egyptian language models for arabic and latin scripts](#). *arXiv preprint arXiv:2507.04569*, arXiv:2507.04569.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. [Deepseekmath: Pushing the limits of mathematical reasoning in open language models](#). *arXiv preprint arXiv:2402.03300*, arXiv:2402.03300.
- Shiqi Shen, Yong Cheng, Zhongjun He, Wei He, Hua Wu, Maosong Sun, and Yang Liu. 2016. [Minimum risk training for neural machine translation](#). In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, pages 1683–1692. Association for Computational Linguistics.
- Serry Sibae, Omer Nacar, Yasser Al-Habashi, Adel Ammar, and Wadii Boulila. 2025. [Shami-mt: A syrian arabic dialect to modern standard arabic bidirectional machine translation system](#). *arXiv preprint arXiv:2508.02268*, arXiv:2508.02268.
- Hao Wang, Linlong Xu, Heng Liu, Yangyang Liu, Xiaohu Zhao, Bo Zeng, Liangying Shao, Longyue Wang, Weihua Luo, and Kaifu Zhang. 2025. [Multi-perspective multi-pair preference optimization for machine translation](#). *arXiv preprint arXiv:2510.13434*, arXiv:2510.13434.
- Sen Yang, Shanbo Cheng, Lu Xu, Jianbing Zhang, and Shujian Huang. 2026. [Grmm: Group relative reward modeling for machine translation](#). *arXiv preprint arXiv:2602.14028*, arXiv:2602.14028.
- Wenjie Yang, Mao Zheng, Mingyang Song, and Zheng Li. 2025. [Ssr-zero: Simple self-rewarding reinforcement learning for machine translation](#). *arXiv preprint arXiv:2505.16637*, arXiv:2505.16637.
- Lily H. Zhang, Hamid Dadkhahi, Mara Finkelstein, Firas Trabelsi, Jiaming Luo, and Markus Freitag. 2025. [Learning from others’ mistakes: Finetuning machine translation models with span-level error annotations](#). In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 74787–74803. PMLR.

A Detailed Per-Language Results

This appendix reports the detailed per-language tables used for the main analysis. These tables are kept outside the main results section to improve readability while preserving transparency.

Language	SFT	DPO	NLLB-600M	Gemma4-E2B	Gemma4-E4B	Gemma4-26B	Qwen-Next	GPT-OSS
Danish	49.10	48.64	48.41	51.81	52.73	<u>53.77</u>	53.14	52.18
Portuguese	48.46	48.06	50.09	52.21	53.18	<u>55.02</u>	54.28	53.94
German	47.26	47.69	44.48	46.80	47.96	<u>49.29</u>	49.26	47.78
Swedish	46.10	45.61	47.35	51.17	52.31	<u>53.99</u>	52.71	53.03
French	45.84	45.88	45.71	47.89	48.75	<u>50.19</u>	49.91	49.16
Romanian	44.65	44.09	45.57	44.63	46.75	<u>48.51</u>	47.94	47.14
Indonesian	43.40	42.59	43.29	46.16	47.39	<u>50.38</u>	48.63	47.67
Turkish	41.24	40.50	39.04	40.80	42.37	<u>44.69</u>	42.92	42.36
Slovak	41.07	40.94	39.12	39.15	42.07	<u>45.26</u>	44.53	42.72
Modern Standard Arabic	41.94	43.91	39.89	40.26	42.10	<u>45.62</u>	44.57	42.30
Egyptian Arabic	27.75	29.80	<u>30.45</u>	26.91	28.36	29.83	28.96	26.91

Table 9: FLORES-200 xx→English results.

Language	DPO model	Gemma4-E2B	Gemma4-E4B	Gemma4-26B
Afrikaans	54.72	51.58	54.62	<u>58.59</u>
Maltese	49.09	32.98	39.93	<u>47.83</u>
Swedish	<u>45.55</u>	45.44	47.05	<u>49.10</u>
Norwegian Bokmål	45.22	44.18	46.25	<u>47.76</u>
Portuguese	44.71	41.98	43.43	<u>45.68</u>
Indonesian	43.79	42.58	43.71	<u>46.72</u>
Filipino	43.17	43.28	45.01	<u>49.38</u>
Galician	43.14	37.38	40.14	<u>43.44</u>
Dutch	42.89	39.65	41.57	<u>43.31</u>
Arabic	<u>37.75</u>	30.41	32.95	37.43

Table 10: NTREX xx→en results

Language	DPO model	Gemma4-E2B	Gemma4-E4B	Gemma4-26B
Mexican Spanish	43.07	44.47	46.19	47.48
Norwegian	42.65	45.37	47.25	<u>49.90</u>
Swedish	42.12	43.65	45.66	<u>47.87</u>
Italian	41.27	41.51	43.08	<u>45.19</u>
Danish	41.02	42.20	43.70	<u>45.00</u>
Romanian	40.60	40.17	42.48	<u>45.43</u>
Filipino	38.87	41.51	42.32	<u>45.27</u>
European Portuguese	38.69	38.31	39.25	<u>40.53</u>
Greek	38.35	39.49	41.95	<u>44.73</u>
Modern Standard Arabic	35.05	31.91	33.72	36.01
Egyptian Arabic	<u>24.91</u>	21.80	22.64	24.49

Table 11: WMT24++ xx→en results

Source	Our model	NLLB-600M	Gemma4-E2B	Gemma4-E4B	Gemma4-26B
English	41.87	34.48	42.04	42.42	<u>43.66</u>
Portuguese	33.42	27.85	33.54	35.16	<u>37.20</u>
German	32.40	26.60	31.80	33.42	<u>34.82</u>
Romanian	32.08	26.68	31.65	33.57	<u>35.34</u>
Turkish	30.39	23.76	28.06	29.86	<u>32.81</u>
Czech	30.01	23.66	28.32	30.39	<u>33.33</u>

Table 12: FLORES-200 xx→MSA results.

Source	Our model	NLLB-600M	Gemma4-E2B	Gemma4-E4B	Gemma4-26B
English	<u>30.02</u>	24.85	28.46	29.40	29.80
Danish	<u>24.92</u>	19.06	23.83	24.89	15.70
German	25.07	18.81	16.88	17.73	25.41
Indonesian	<u>23.77</u>	16.59	22.32	23.39	18.03
Persian	22.99	17.56	14.58	22.49	23.44
Turkish	23.73	17.82	21.26	23.31	<u>24.53</u>

Table 13: FLORES-200 xx→ECA results.

Source	Our model	Gemma4-E2B	Gemma4-E4B	Gemma4-26B
English	34.99	35.48	<u>35.53</u>	35.32
Afrikaans	29.41	26.08	29.41	<u>31.45</u>
Maltese	29.20	20.70	24.53	<u>29.60</u>
Portuguese	28.54	28.04	29.07	<u>29.58</u>
Italian	28.01	27.99	28.43	<u>29.25</u>
Spanish	27.87	27.19	27.83	<u>28.68</u>

Table 14: NTREX-128 xx→MSA results.

Source	Gemma4-E2B	Gemma4-E4B	Gemma4-26B	Our model
Arabic	24.06	24.86	25.96	24.65
English	20.16	22.42	<u>25.72</u>	<u>26.03</u>
French	18.42	20.25	<u>23.34</u>	21.44
Danish	16.57	17.83	<u>22.34</u>	19.54
Romanian	18.21	20.02	<u>23.91</u>	21.33

Table 15: WMT24++ xx→ECA results.

B Supported Languages

We provide support for the following language-control tokens. Each token is used to indicate the desired target language during training and inference.

Token	Language	Token	Language	Token	Language
__no__	Norwegian	__nr__	South Ndebele	__nso__	Northern Sotho
__nv__	Navajo	__ny__	Chichewa	__om__	Oromo
__os__	Ossetian	__pa__	Punjabi	__pap__	Papiamento
__pl__	Polish	__ps__	Pashto	__pt__	Portuguese
__qu__	Quechua	__ro__	Romanian	__ru__	Russian
__run__	Kirundi	__rw__	Kinyarwanda	__sg__	Sango
__si__	Sinhala	__sk__	Slovak	__sl__	Slovenian
__sm__	Samoan	__sn__	Shona	__sq__	Albanian
__sr__	Serbian	__srn__	Sranan Tongo	__ss__	Swati
__st__	Southern Sotho	__su__	Sundanese	__sv__	Swedish
__sw__	Swahili	__swc__	Congo Swahili	__ta__	Tamil
__te__	Telugu	__tg__	Tajik	__th__	Thai
__ti__	Tigrinya	__tk__	Turkmen	__tl__	Tagalog
__tn__	Tswana	__to__	Tongan	__tpi__	Tok Pisin
__tr__	Turkish	__ts__	Tsonga	__tt__	Tatar
__tum__	Tumbuka	__tw__	Twi	__ty__	Tahitian
__uk__	Ukrainian	__umb__	Umbundu	__ur__	Urdu
__uz__	Uzbek	__ve__	Venda	__vi__	Vietnamese
__war__	Waray	__xh__	Xhosa	__yap__	Yapese
__yo__	Yoruba	__zh__	Chinese	__zu__	Zulu
__arz__	Egyptian Arabic	__en__	English	__ar__	Modern Standard Arabic

Table 16: Supported language-control by our model.

C Qualitative Examples

Figure 1 presents two representative qualitative examples comparing the SFT and DPO outputs.

1 Example 1: Idiomaticity and naturalness		
Source (English)	SFT Output	DPO Output
Stop being a copy cat and try to come up with your own thesis topic for the seminar.	بطل تبقى قطة تقليدية و حاول تعمل موضوع أطروحة خاص بيك للندوة	بلاش تبقى زي الناس الثانية ، وحاول تقدم موضوع أطروحة بنفسك للحلقة الدراسية دي

2 Example 2: Adequacy and meaning preservation		
Source (English)	SFT Output	DPO Output
I wasn't upset because you were late; I was upset because you acted like my time didn't matter.	أنا مش متضايق عشان انت اتأخرت ، أنا متضايق عشان انت بتتصرف زي زمانى مش مهم	أنا ما كنتش متضايق عشان انت اتأخرت ، أنا كنت متضايق عشان انت اتصرفت كأن وقتي مش مهم

Figure 1: Qualitative comparison between SFT and DPO outputs on two English-to-Egyptian-Arabic examples. The DPO model produces more natural and semantically accurate translations.

D LLM Baseline Prompting and Decoding Protocol

This appendix describes the prompting and decoding protocol used for the LLM baselines reported in the main results. For all LLM baselines, we used the same fixed translation prompt across all evaluated language directions. The prompt template was:

Translate the following sentence from {source_lang} to {target_lang}: {text}
output only the translated text.

Here, {source_lang} and {target_lang} were replaced with the corresponding source and target language names, and {text} was replaced with the input sentence to be translated. The same prompt structure was used for all baseline models to avoid model-specific prompt tuning.

For generation, we used each baseline model with its recommended temperature setting. Apart from this model-specific temperature choice, the evaluation protocol was kept consistent across baselines. Each model was instructed to output only the translated sentence, and the generated translation text was used for spBLEU evaluation. No reference translations were provided to the LLM baselines during generation.

E DPO Preference-Data Generation

E.1 Motivation and Construction Overview

The preference data construction was motivated by recent work applying Group Relative Policy Optimization to encoder-decoder machine translation (Garcia-Estrada et al., 2026). That work highlights the difficulty of obtaining sufficiently discriminative scalar rewards for Arabic translation, particularly when candidate differences involve morphology, lexical choice, or other fine-grained linguistic phenomena.

Instead of relying on a scalar reward during online optimization, we construct explicit preferred–rejected translation pairs and optimize them using DPO.

The construction process consists of two stages. First, Gemini 3 Flash generates a multi-parallel record containing an English sentence and semantically aligned translations in 19 additional language varieties. Second, for each selected source–target direction, the corresponding translation generated by Gemini is used as the preferred candidate y^+ , while the output produced by our supervised fine-tuned (SFT) translation checkpoint for the same source sentence and target-language condition is used as the rejected candidate y^- . Therefore, no additional LLM prompt is used to generate the rejected translation: it is obtained directly from the SFT model that DPO is intended to improve.

E.2 Languages

Each generated multi-parallel record contains the following 20 language varieties:

English, Modern Standard Arabic, Egyptian Arabic, French, German, Spanish, Italian, Polish, Portuguese, Simplified Chinese, Korean, Japanese, Hindi, Russian, Ukrainian, Turkish, Urdu, Romanian, Vietnamese, and Dutch.

The use of multi-parallel records allows the same underlying meaning to be used across multiple source–target directions while maintaining a common semantic reference.

E.3 Domain Control

The generation script supplies a randomized domain to Gemini for each record. The generated examples cover multiple general and specialized domains, including sports, education, law, medicine, and climate-related topics.

Domain control serves two purposes. First, it prevents the preference data from being limited to generic conversational sentences. Second, it introduces domain-specific lexical and register requirements. For example, legal sentences may require distinctions involving due process, contractual remedies, procedural safeguards, or fundamental rights, whereas sports examples may involve abstract descriptions of psychological resilience, athletic performance, and collective effort.

E.4 Difficulty Control

Difficulty is controlled through a fixed natural-language rubric in the generation prompt rather than through a separate automatic metric or a human-assigned numerical score. Gemini is instructed to generate a semantically coherent but linguistically challenging sentence and to provide a textual explanation in the `difficulty_reason` field.

The difficulty rationale identifies the linguistic properties that make the example challenging to translate. Across the generated records, these properties include:

- nested subordinate, conditional, or causal structures;
- abstract or philosophical concepts;
- nuanced lexical distinctions;
- formal and domain-specific terminology;
- modality and legal relationships;
- idiomatic or non-literal phrasing;
- register differences across languages; and
- the need to restructure syntax naturally rather than translate word by word.

E.5 Generation Prompt

The following prompt template describes the instructions used to generate the multi-parallel records.

You are an expert multilingual translation-data generator.
 Requested field: {field}
 Generate exactly one original English sentence related to the requested field. The sentence must be natural and semantically coherent, but linguistically challenging to translate.
 The difficulty should arise from one or more of the following: complex causal, conditional, or subordinate structures; abstract concepts; nuanced lexical distinctions; idiomatic or non-literal phrasing; domain-specific terminology; formal register; or syntactic structures that must be reorganized naturally across languages.
 Do not make the sentence difficult merely by adding obscure words or unnecessary length. It should express a meaningful idea appropriate to the requested field.
 Translate the sentence into all of the following language varieties:
 English, Modern Standard Arabic, Egyptian Arabic, French, German, Spanish, Italian, Polish, Portuguese, Simplified Chinese, Korean, Japanese, Hindi, Russian, Ukrainian, Turkish, Urdu, Romanian, Vietnamese, and Dutch.
 Requirements:
 1. Preserve the complete meaning of the English sentence. 2. Produce natural translations rather than word-for-word renderings. 3. Preserve the intended register and domain terminology. 4. Keep Modern Standard Arabic and Egyptian Arabic distinct. 5. The Egyptian Arabic translation must use natural Egyptian vocabulary, grammar, and phrasing rather than simplified Modern Standard Arabic. 6. Do not omit or introduce information. 7. Explain briefly why the sentence is difficult to translate.
 Return only one JSON object using the following structure:

```
{ "field": "{field}", "difficulty_reason": "A concise explanation of the linguistic difficulty.",
  "translations": { "english": "...", "arabic": "...", "egyptian_arabic": "...", "french": "...",
    "german": "...", "spanish": "...", "italian": "...", "polish": "...", "portuguese": "...",
    "chinese_simplified": "...", "korean": "...", "japanese": "...", "hindi": "...", "russian": "...",
    "ukrainian": "...", "turkish": "...", "urdu": "...", "romanian": "...", "vietnamese": "...",
    "dutch": "..." } }
```

E.6 Preference-Pair Formation

After generation, each multi-parallel record can be converted into one or more translation directions. For a selected source language L_s and target language L_t , we construct the preference pair as follows:

1. The sentence in L_s is selected as the source x .
2. The corresponding Gemini translation in L_t is assigned as the preferred candidate y^+ .
3. The source sentence is translated into L_t using the SFT checkpoint with the appropriate target-language control token.
4. The SFT output is assigned as the rejected candidate y^- .

This construction directly targets the failure modes observed in the SFT checkpoint. Depending on the translation direction, the difference between the preferred and rejected candidates may involve literal translation, lexical ambiguity, inappropriate register, unnatural Egyptian Arabic phrasing, morphology, or domain terminology. Figure 2 illustrates a representative English-to-Egyptian-Arabic preference pair. The final DPO records contain the source sentence, preferred translation, rejected translation, source-language token, target-language token, and translation direction. Malformed and duplicate records are removed during filtering. Pairs for which the preferred and rejected candidates are effectively indistinguishable are also excluded.

Field: sports

Source:

The more athletes fixate on the outcome rather than the process, the more likely they are to falter when the pressure of high-stakes competition demands absolute presence of mind.

✓ **Chosen:**

كل ما الرياضيين ركزوا في النتيجة أكثر من الأداء نفسه ، كل ما بقى أسهل إنهم
يقعوا لما ضغط المسابقات الكبيرة يحتاج منهم تركيز وحضور ذهني عالي .

✗ **Rejected:**

كل ما الرياضيين بيركزوا على النتيجة بدل العملية ، كل ما بيكونوا ممكنين
أكثر إنهم يقعوا تحت ضغط المنافسة العالية اللي بتتطلب وجود مطلق للعقل .

Figure 2: Representative DPO preference example from the sports domain. The highlighted spans show natural Egyptian Arabic phrasing in the chosen translation and literal or awkward phrasing in the rejected SFT output.